

Topic B19

Human influences on ecosystems

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

B20.1	Food supply
B20.2	Habitat destruction
B20.3	Pollution
B20.4	Conservation

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Food production

Agricultural yield can increase through fertilisers, irrigation, pesticides, selective breeding, machinery and controlled environments. Intensive systems improve output but may reduce biodiversity, pollute water, select resistant pests and affect animal welfare. Sustainable approaches balance yield with long-term soil and ecosystem health.

Examiner insight

Evaluation needs benefits, environmental costs and a justified judgement.

2. Habitat destruction

Deforestation removes habitats, reduces biodiversity, exposes soil to erosion and decreases carbon storage. Land is cleared for timber, agriculture, mining and settlement. Fragmentation isolates populations, reduces gene flow and increases extinction risk.

Examiner insight

Connect the human action to habitat change, population effect and ecosystem consequence.

3. Pollution

Untreated sewage and fertiliser cause eutrophication. Persistent pesticides can bioaccumulate within organisms and biomagnify up food chains. Plastics injure animals and persist; sulfur dioxide contributes to acid rain; greenhouse gases alter climate.

Examiner insight

Bioaccumulation occurs within an organism; biomagnification increases concentration across trophic levels.

4. Conservation

Conservation maintains biodiversity through protected areas, legal controls, habitat restoration, captive breeding, seed banks and sustainable harvesting. Small populations may need genetic management. Successful conservation includes monitoring and cooperation with local communities.

Examiner insight

Conservation is not simply preventing all human use; sustainable management can protect ecosystems and livelihoods.

Worked application

Situation: A forest is divided by roads into small fragments.

Prediction: Local extinction risk increases.

Reason: Populations become smaller and isolated, reducing gene flow and access to resources.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic B19

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Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Which statement best defines biodiversity? [1]

- A the variety of genes, species and ecosystems in an area
- B build-up of a substance within an organism over time
- C increase in pollutant concentration at successive trophic levels
- D management and protection of biodiversity and resources

2. Which statement best defines bioaccumulation? [1]

- A increase in pollutant concentration at successive trophic levels
- B build-up of a substance within an organism over time
- C management and protection of biodiversity and resources
- D a resource used at a rate that allows replacement and long-term availability

3. Which statement best defines biomagnification? [1]

- A management and protection of biodiversity and resources
- B a resource used at a rate that allows replacement and long-term availability
- C increase in pollutant concentration at successive trophic levels
- D the variety of genes, species and ecosystems in an area

4. Which statement best defines conservation? [1]

- A a resource used at a rate that allows replacement and long-term availability
- B the variety of genes, species and ecosystems in an area
- C build-up of a substance within an organism over time
- D management and protection of biodiversity and resources

5. Which statement best defines sustainable resource? [1]

- A a resource used at a rate that allows replacement and long-term availability
- B the variety of genes, species and ecosystems in an area
- C build-up of a substance within an organism over time
- D increase in pollutant concentration at successive trophic levels

6. Which statement about food production is correct? [1]

- A The number of individuals of one species only.
- B Agricultural yield can increase through fertilisers, irrigation, pesticides, selective breeding, machinery and controlled environments.
- C Increase between trophic levels.
- D Rapid decomposition of a pollutant.

7. Which statement about habitat destruction is correct? [1]
- A Increase between trophic levels.
 - B Rapid decomposition of a pollutant.
 - C Deforestation removes habitats, reduces biodiversity, exposes soil to erosion and decreases carbon storage.
 - D Complete removal of people from every habitat.
8. Which statement about pollution is correct? [1]
- A Rapid decomposition of a pollutant.
 - B Complete removal of people from every habitat.
 - C Any resource used for profit.
 - D Untreated sewage and fertiliser cause eutrophication.
9. Which statement about conservation is correct? [1]
- A Conservation maintains biodiversity through protected areas, legal controls, habitat restoration, captive breeding, seed banks and sustainable harvesting.
 - B Complete removal of people from every habitat.
 - C Any resource used for profit.
 - D The number of individuals of one species only.
10. Which prediction is most appropriate when a forest is divided by roads into small fragments? [1]
- A The number of individuals of one species only.
 - B Rapid decomposition of a pollutant.
 - C Local extinction risk increases.
 - D Any resource used for profit.

Section B · Structured questions

11 Biologists use precise vocabulary when studying human influences on ecosystems.

(a) Define biodiversity.

[2]

(b) Define bioaccumulation.

[2]

(c) Distinguish biomagnification from the incorrect idea: “rapid decomposition of a pollutant”.

[2]

(d) State two important features of food production.

[2]

[Total: 8]

12 A forest is divided by roads into small fragments.

(a) State the most likely biological outcome.

[2]

(b) Explain the outcome using a linked biological mechanism.

[3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records relative concentration under three conditions.

condition	relative concentration
water	1
small fish	18
fish-eating bird	140

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to compare plant diversity in managed and unmanaged grassland using quadrats.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of human influences on ecosystems to an unfamiliar context.

(a) Define conservation. [2]

(b) Explain the biological importance of pollution. [3]

(c) Correct this statement: “any resource used for profit”.

[2]

(d) State one further question that could be investigated.

[1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. The variety of genes, species and ecosystems in an area.
- 2 B. Build-up of a substance within an organism over time.
- 3 C. Increase in pollutant concentration at successive trophic levels.
- 4 D. Management and protection of biodiversity and resources.
- 5 A. A resource used at a rate that allows replacement and long-term availability.
- 6 B. Evaluation needs benefits, environmental costs and a justified judgement.
- 7 C. Connect the human action to habitat change, population effect and ecosystem consequence.
- 8 D. Bioaccumulation occurs within an organism; biomagnification increases concentration across trophic levels.
- 9 A. Conservation is not simply preventing all human use; sustainable management can protect ecosystems and livelihoods.
- 10 C. Populations become smaller and isolated, reducing gene flow and access to resources.

Solutions · Structured questions

11. (a) the variety of genes, species and ecosystems in an area [2]. (b) build-up of a substance within an organism over time [2]. (c) increase in pollutant concentration at successive trophic levels; not rapid decomposition of a pollutant [2]. (d) Credit any two accurate features from the guide: Agricultural yield can increase through fertilisers, irrigation, pesticides, selective breeding, machinery and controlled environments. Intensive systems improve output but may reduce biodiversity, pollute water, select resistant pests and affect animal welfare. Sustainable approaches balance yield with long-term soil and ecosystem health. [2].

12. (a) Local extinction risk increases [2]. (b) Populations become smaller and isolated, reducing gene flow and access to resources [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a) $140 - 1 = 139$ [2]. (b) The values change from 1 to 140, with the highest value 140. [2]. (c) Persistent pesticide becomes more concentrated at higher trophic levels through biomagnification [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) Management type [1]. (b) Species richness or diversity index [1]. (c) Same quadrat size, effort, season and area [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Randomise many quadrats and identify species consistently [1].

15. (a) management and protection of biodiversity and resources [2]. (b) Award linked points from: Untreated sewage and fertiliser cause eutrophication. Persistent pesticides can bioaccumulate within organisms and biomagnify up food chains. Plastics injure animals and persist; sulfur dioxide contributes to acid rain; greenhouse gases alter climate. [3]. (c) Correct statement: sustainable resource means a resource used at a rate that allows replacement and long-term availability [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.